

# 2023 Bi H2 Q8

## Section: Metabolism and Survival

### Topic: Metabolism and Adverse Conditions

#### Question Summary

This question looks at the breathing rate of desert tortoises through the year, how this relates to hibernation, and compares it with metabolic adaptations in other animals such as sugar gliders.

#### Worked Solution

(a)(i) Breathing rates:

- January = 2.8
- March = 14.0
- October = 16.8

Simplify the ratio 2.8 : 14.0 : 16.8.

Divide each by 2.8 → 1 : 5 : 6.

■ Answer: 1 : 5 : 6

(a)(ii) Lowest breathing rates occur in January, February, November, and December — indicating hibernation.

■ Answer: These are the months when tortoises were hibernating.

(b) When there is a drought, desert tortoises reduce their heart and breathing rates. This is a dormancy response called

aestivation (or aestivation response).

Explanation: By slowing metabolism, tortoises conserve energy and water, helping them survive when conditions are hot and dry.

■ Answer: Aestivation — reduces metabolism to conserve energy and water.

(c)(i) Sugar gliders lower their metabolic rate each day — this short-term, daily dormancy is called torpor.

■ Answer: Torpor

(c)(ii) Reducing metabolic rate during the day saves energy when the animal is resting and not feeding.

■ Answer: Energy conservation during inactivity

## **Revision Tips**

- Hibernation: Long-term dormancy during cold or low-food periods.
- Aestivation: Dormancy during hot or dry conditions.
- Torpor: Short-term daily drop in metabolic rate (seen in small mammals and birds).
- All three reduce metabolism to conserve energy when survival conditions are poor.
- Remember: Hibernation → cold, Aestivation → dry, Torpor → daily cycle.

# Annual breathing rate of tortoise

